

VELAGAPUDI RAMAKRISHNA SIDDHARTHA ENGINEERING COLLEGE

Department of Artificial Intelligence and Data Science

RE PLASTIC INNOVATIONS

Salesforce-Based Inventory Management System

PROJECT DOCUMENTATION

Salesforce CRM and Inventory Automation

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Salesforce Platform Development Project

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1 Project Overview

1.1 Introduction

Re Plastic Innovations is a Salesforce-based inventory management solution designed to manage recycled products, customer orders, stock levels, and restocking activities in a centralized Salesforce environment.

The project combines Salesforce configuration, formula fields, validation rules, scheduled Flow automation, Apex classes, Apex triggers, email notification logic, and functional testing.

The system is designed around a practical inventory scenario in which customer orders can affect product stock. When available stock is insufficient to satisfy an order, the system automatically creates a restock request for the required quantity. Once the restock request is approved, Apex updates the product stock and sends an email notification.

Core workflow:

Order Creation → Stock Verification → Restock Request → Approval → Stock Update → Email Notification

1.2 Technology Used

Table 1: Technology and Salesforce Features Used

Area	Implementation
Platform	Salesforce Platform
Data Management	Custom Objects and Custom Fields
Relationships	Lookup Relationships
Business Logic	Formula Fields and Validation Rules
Automation	Scheduled-Triggered Flow
Programming	Apex Classes and Apex Triggers
Communication	Apex Email Notification using Messaging.SingleEmailMessage
Testing	Apex Test Class and Code Coverage
User Interface	Salesforce Lightning Experience

1.3 System Overview

The completed Salesforce application provides a centralized environment for managing recycled products and their inventory-related transactions.

The main operational sequence is:

1. A recycled product is maintained with its stock level and threshold.
2. A customer order is created for the product.
3. Apex checks the available stock.
4. If stock is sufficient, the stock is reduced by the ordered quantity.

5. If stock is insufficient, a Restock Request is automatically created.
6. The request initially remains in Pending status.
7. When the request changes from Pending to Approved, the stock is increased.
8. An email notification is generated confirming the approval.
9. A scheduled Flow identifies products whose stock is below their threshold and creates a high-priority Task.

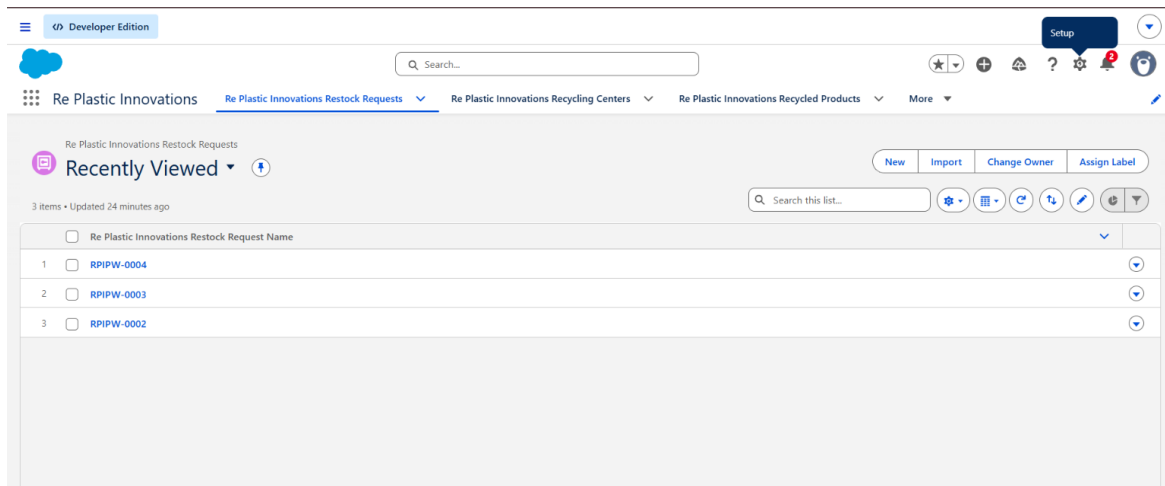


Figure 1: Re Plastic Innovations Salesforce application overview

2 Objectives and Implementation Scope

2.1 Project Objectives

The major objectives of the project are:

- Create a structured Salesforce data model for recycled products, orders, and restock requests.
- Maintain accurate inventory information using Salesforce fields and business rules.
- Prevent invalid order quantities through a validation rule.
- Calculate a product stock status using a formula field.
- Automatically identify products whose stock falls below the defined threshold.
- Create a Task automatically for low-stock products.
- Automate stock processing using Apex.
- Automatically create Restock Requests when an order quantity exceeds available stock.
- Update stock when a Restock Request is approved.
- Notify the user through email after restock approval.
- Test the implementation using Apex test methods and functional Salesforce records.

2.2 Implementation Scope

Table 2: Project Implementation Scope

Area	Implementation	Evidence
Data Model	Custom objects and relationships	Object Manager / Schema Builder
Formula Logic	Stock status formula	Recycled Product record
Data Validation	Order quantity validation	Validation Rule and error screen
Automation	Scheduled low-stock Flow	Flow Builder and Task
Apex Development	InventoryManager and triggers	Developer Console
Email Automation	Restock approval email	Email inbox
Testing	Apex test class and functional testing	Test Results / Coverage

3 Requirement Analysis and Data Model

3.1 Custom Objects

The project uses Salesforce custom objects to represent the main entities involved in inventory management.

Table 3: Re Plastic Innovations Custom Objects

Custom Object	Important Fields	Purpose
Recycled Product	Stock Level, Threshold, Price	Maintains recycled product and inventory information.
Order	Customer, Recycled Product, Quantity, Delivery Date	Stores customer order information.
Restock Request	Recycled Product, Requested Quantity, Status	Stores stock replenishment requests generated by the system.
Plastic Waste	Project-specific waste information	Represents plastic waste information maintained in the Salesforce application.
Recycling Center	Recycling center information	Represents recycling center records associated with the business process.

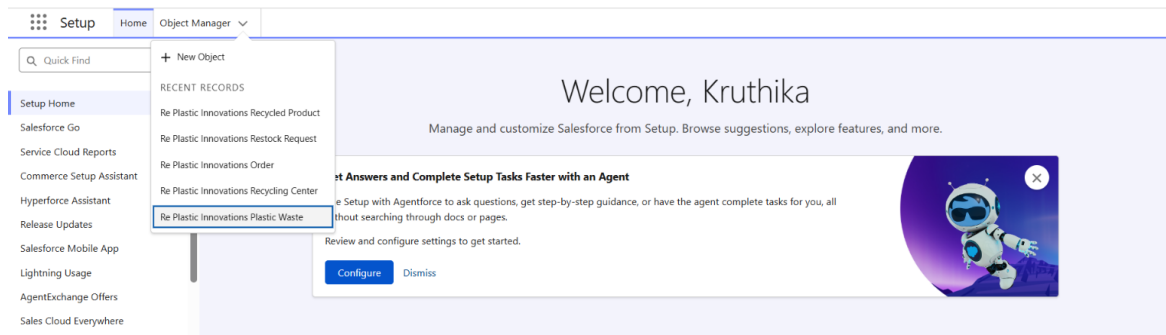


Figure 2: Custom objects configured for the Re Plastic Innovations application

3.2 Recycled Product Object

The Recycled Product object is the central inventory object.

Important fields include:

- **Product Name** – identifies the recycled product.
- **Stock Level** – represents the current quantity available.
- **Threshold** – represents the minimum desired stock level.
- **Price** – represents the product price.
- **Stock Low On Product** – formula field that identifies whether stock requires replenishment.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Price	Price_c	Currency(18, 0)		
Re Plastic Innovations Recycled Product Name	Name	Text(80)		✓
Stock Low On Product	Stock_Low_On_Product_c	Formula (Text)		
Stock Level	Stock_Level_c	Number(18, 0)		
Threshold	Threshold_c	Number(18, 0)		

Figure 3: Recycled Product fields and configuration

3.3 Order Object

The Order object records customer orders.

Important fields include:

- Customer
- Recycled Product
- Quantity
- Delivery Date

The Order object is also used by the Apex stock-processing automation.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Customer	Customer__c	Lookup(Account)		✓
Delivery Date	Delivery_Date__c	Date		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Quantity	Quantity__c	Number(18, 0)		
Re Plastic Innovations Order Name	Name	Auto Number		✓
Re Plastic Innovations Recycled Product	Re_Plastic_Innovations_Recycled_Product__c	Lookup(Re Plastic Innovations Recycled Product)		✓

Figure 4: Re Plastic Innovations Order fields and relationships

3.4 Restock Request Object

The Restock Request object represents an inventory replenishment request.

Important fields include:

- Recycled Product
- Requested Quantity
- Status

The normal lifecycle implemented in the project is:

Pending → Approved

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Re Plastic Innovations Recycled Product	Re_Plastic_Innovations_Recycled_Product__c	Lookup(Re Plastic Innovations Recycled Product)		✓
Re Plastic Innovations Restock Request Name	Name	Auto Number		✓
Requested Quantity	Requested_Quantity__c	Number(18, 0)		
Status	Status__c	Picklist		

Figure 5: Restock Request fields and configuration

3.5 Object Relationships

The Order object references the Recycled Product object, while the Restock Request also references the Recycled Product object.

Table 4: Important Object Relationships

Child Object	Related Object	Purpose
Order	Recycled Product	Identifies the product being ordered.
Restock Request	Recycled Product	Identifies the product requiring replenishment.
Order	Customer	Associates an order with the customer.

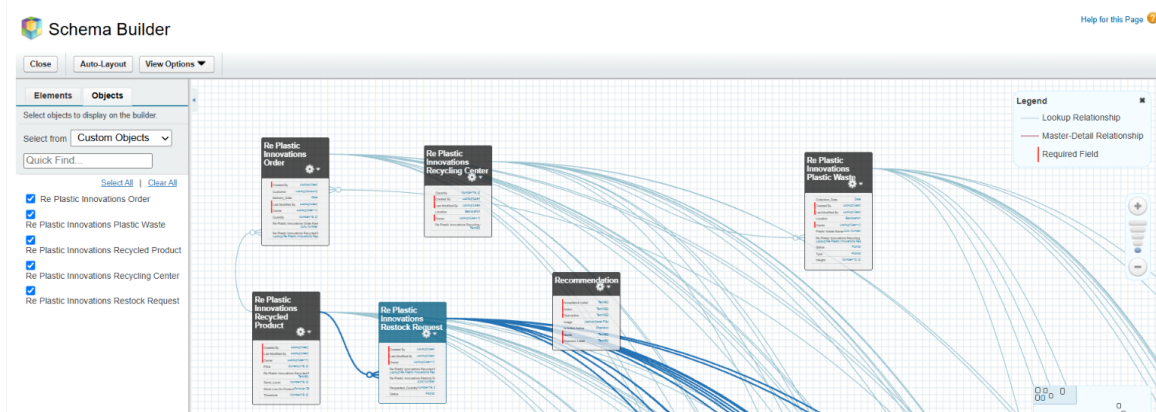


Figure 6: Schema Builder showing the project data relationships

4 Salesforce Configuration and Data Security

4.1 Lightning Application

A dedicated Re Plastic Innovations Lightning application is used to provide access to the configured Salesforce objects.

The application provides navigation to relevant records including recycled products and restock requests.

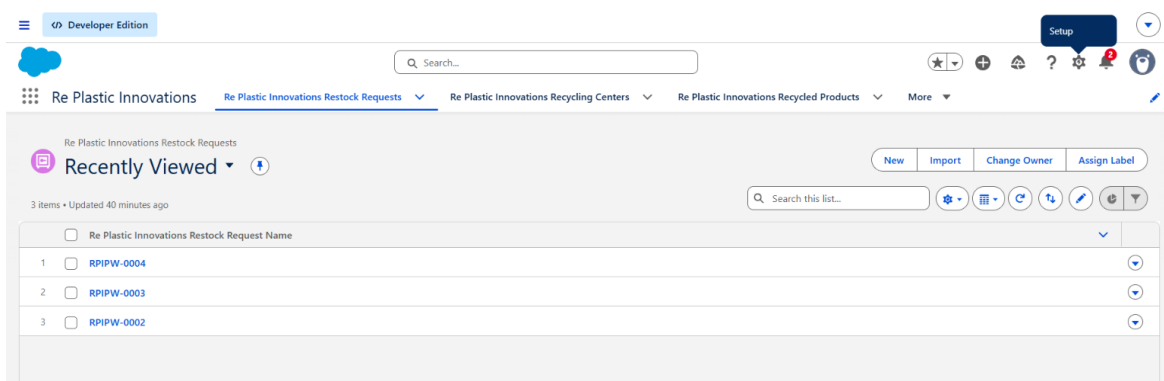


Figure 7: Re Plastic Innovations Lightning application and navigation

4.2 Users and Access Configuration

Salesforce security configuration can be documented using the configured profiles, roles, users, and object-level permissions available in the project org.

Table 5: Security Configuration Evidence

Security Component	Purpose
Profiles	Define base user permissions.
Roles	Represent organizational access structure.
Users	Provide individual Salesforce access.
Object Permissions	Control access to configured objects.
Sharing Configuration	Controls record-level access where configured.

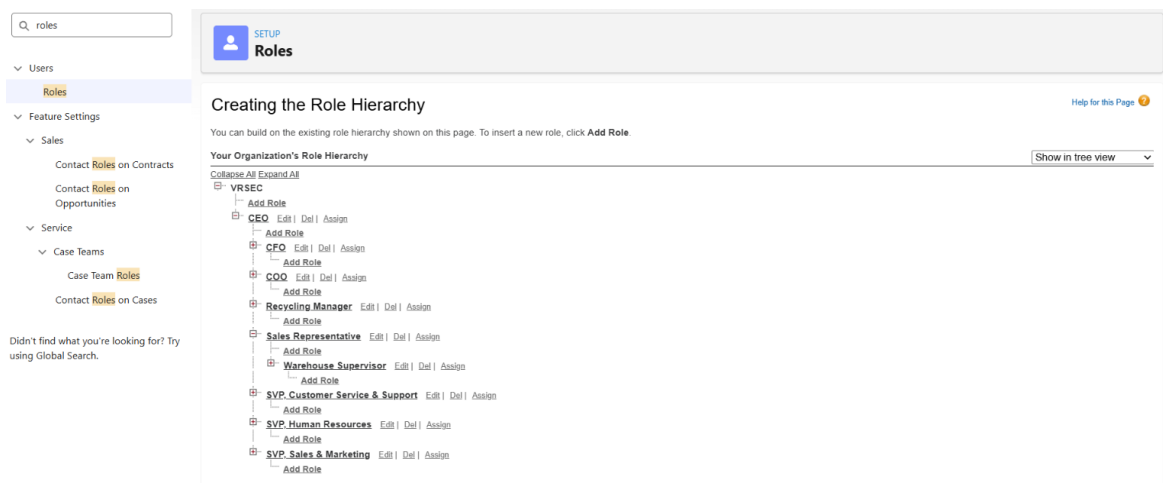


Figure 8: Salesforce role hierarchy

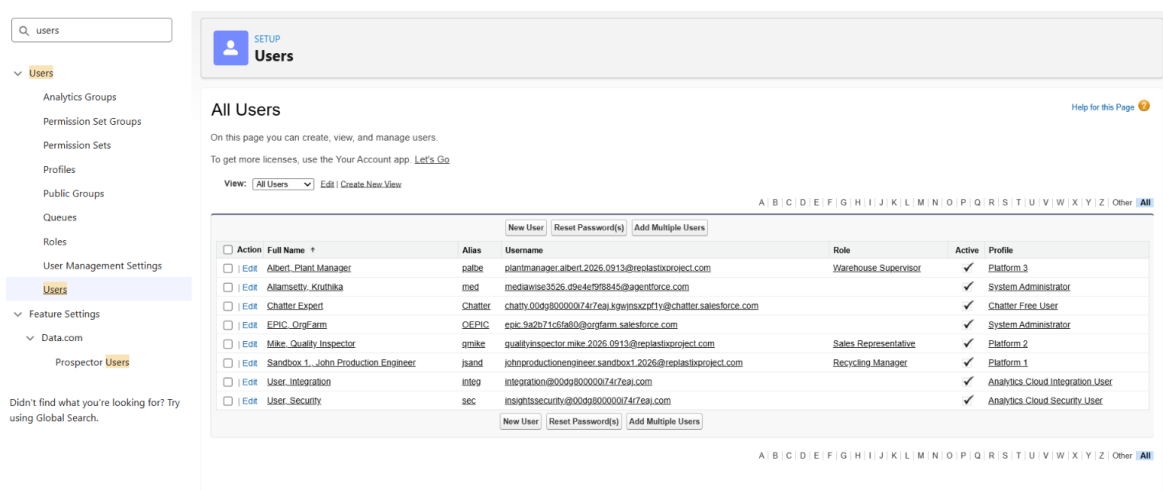


Figure 9: Profiles and project users

5 Data Quality and Business Rules

5.1 Stock Status Formula Field

A formula field named:

Stock Low On Product

was created on the Recycled Product object.

The formula used is:

```

1 IF (
2     Stock_Level__c < Threshold__c ,
3     "Low Stock - Restock Needed",
4     "Sufficient Stock"
5 )

```

The formula dynamically displays the stock status based on the current stock level and threshold.

Table 6: Stock Status Formula Logic

Stock Level	Threshold	Formula Result
Less than Threshold	100	Low Stock - Restock Needed
Greater than or equal to Threshold	100	Sufficient Stock

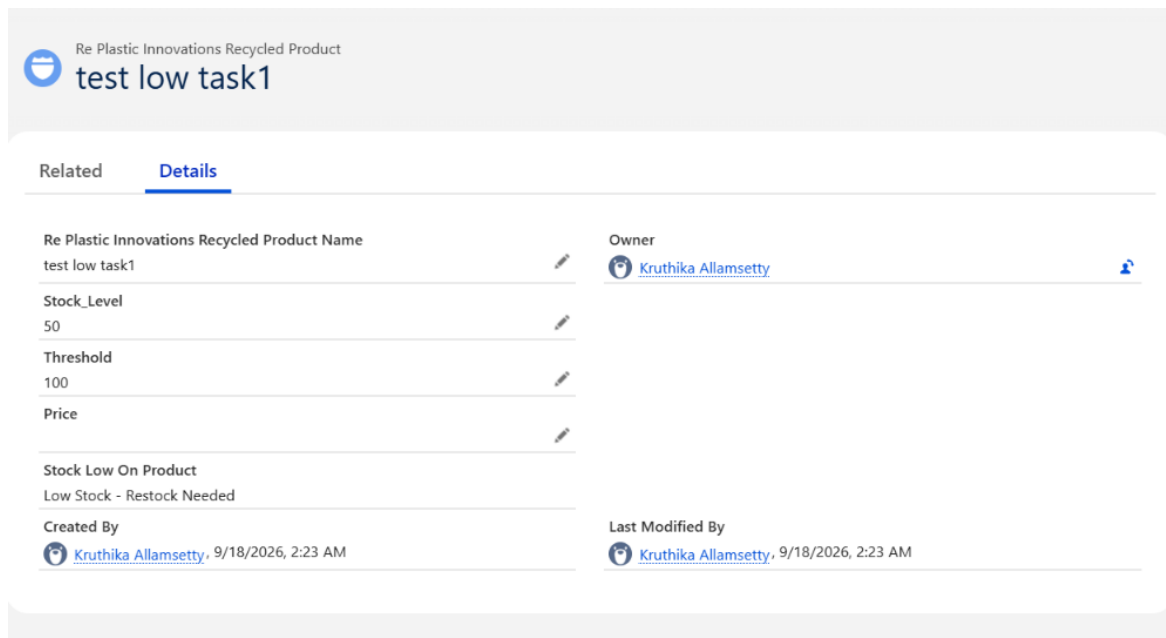


Figure 10: Stock Low On Product formula result on a Recycled Product record

5.2 Order Quantity Validation Rule

A validation rule named:

Check_Quantity_Not_Zero

was configured on the Order object.

Formula:

```
1 Quantity__c <= 0
```

Error Message:

Quantity must be greater than zero.

This prevents an order from being saved when the quantity is zero or negative.

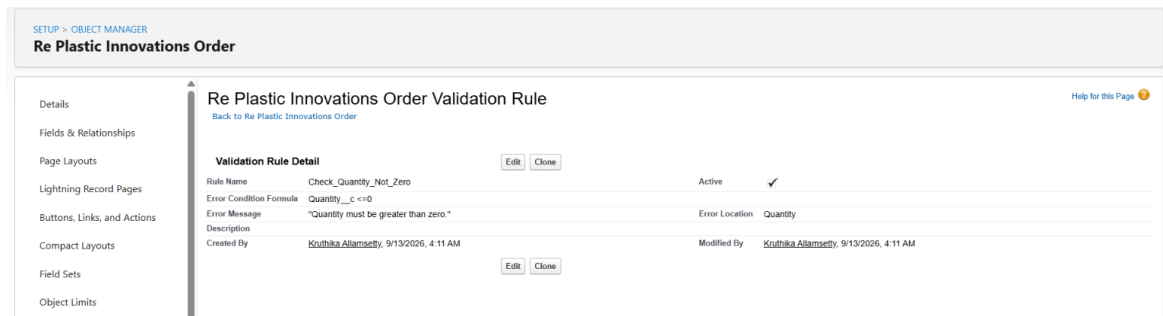


Figure 11: Order quantity validation rule and validation error

5.3 Collection Date Validation

A validation rule named:

Future_Date_Collection

was configured for the Plastic Waste object.

The implemented condition is:

```
1 Collection_Date__c > TODAY()
```

The validation message prevents an invalid future collection date from being saved.

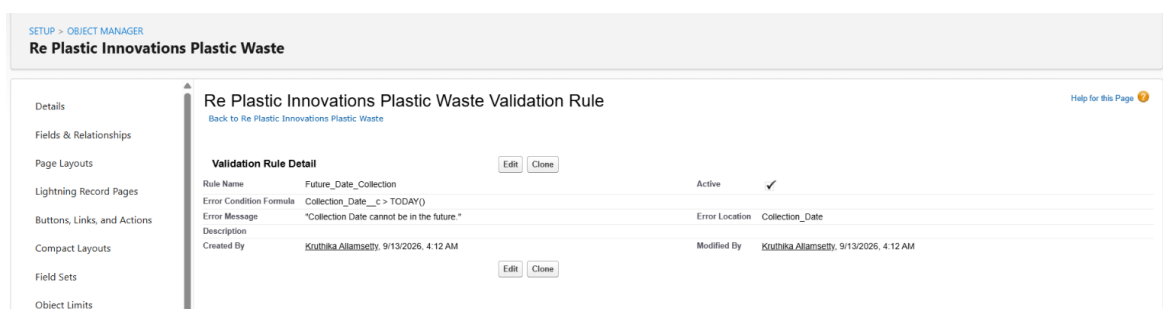


Figure 12: Collection date validation rule

6 Scheduled Flow Automation

6.1 Stock Level Is Low Flow

A Scheduled-Triggered Flow named:

Stock Level Is Low

was created to automatically identify recycled products whose stock level is below the configured threshold.

The Flow runs on the:

Re Plastic Innovations Recycled Product

object.

6.2 Flow Configuration

The Flow follows this sequence:

Schedule → Decision 1 → Create Task → End

The main Decision condition is:

```
1 $Record.Stock_Level__c
2 Less Than
3 $Record.Threshold__c
```

When the condition is true, a Task is created.

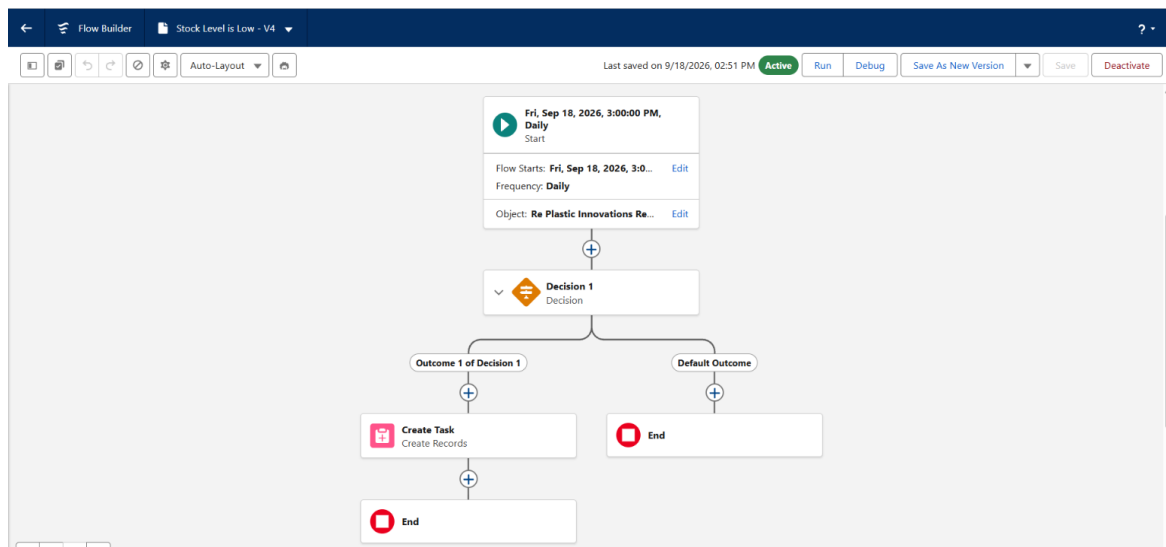


Figure 13: Stock Level Is Low Scheduled-Triggered Flow

6.3 Create Task Configuration

The Flow creates a Salesforce Task with the following values:

Table 7: Low Stock Task Configuration

Task Field	Configured Value
Assigned To ID	Record Owner ID
Related To ID	Recycled Product Record ID
Priority	High
Subject	Please Look in this Stock Is Low Fill The Stock ASAP
Status	In Progress

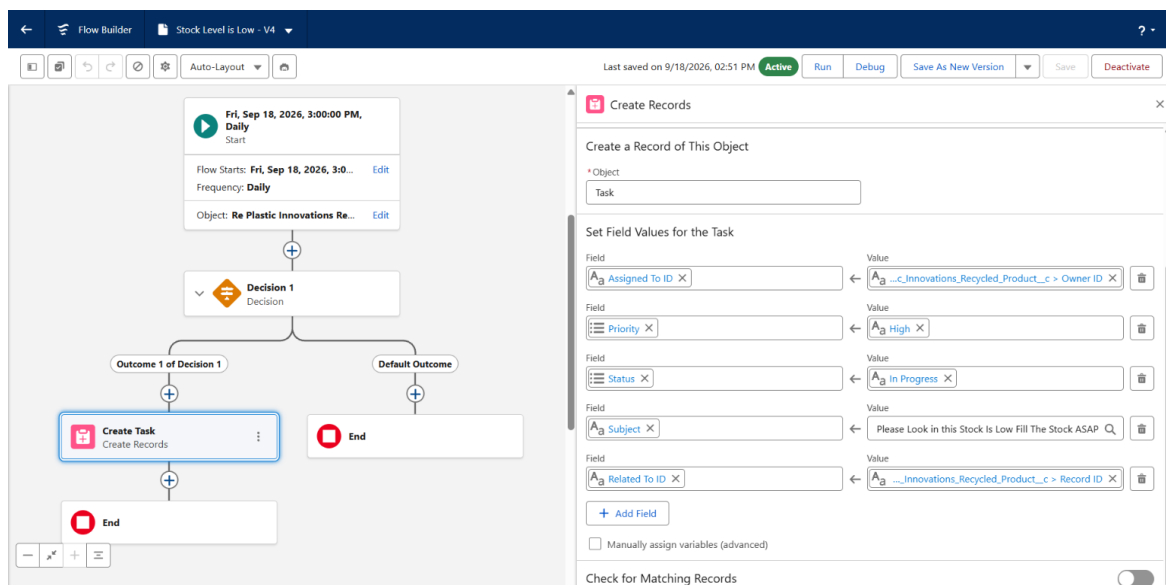


Figure 14: Create Task configuration in Flow Builder

6.4 Flow Test Result

A recycled product was tested with:

$$\text{Stock Level} = 50 \quad \text{Threshold} = 100$$

Because the stock level was below the threshold, the Flow created the configured Task.

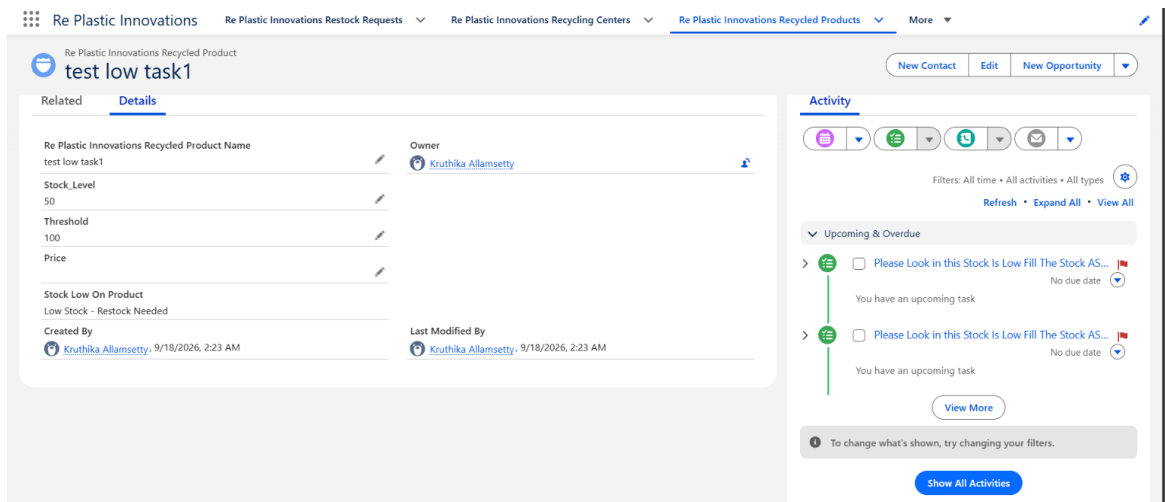


Figure 15: Automatically created low-stock Task displayed under Upcoming and Overdue

Automation Result: The scheduled Flow successfully identified a low-stock product and created a high-priority Task associated with the recycled product record.

7 Apex Development and Trigger Automation

7.1 Apex Architecture

Apex was used for transaction-level inventory processing.

The implementation contains:

- **InventoryManager** – reusable business logic for stock processing.
- **UpdateStockAfterOrder** – processes stock after an Order is inserted.
- **UpdateStockAfterRestockApproval** – processes approved Restock Requests.
- **EmailNotificationHelper** – sends the restock approval email.
- **InventoryManagerTest** – verifies the Apex implementation.

7.2 InventoryManager Class

The InventoryManager class contains two main operations.

Table 8: InventoryManager Responsibilities

Method	Responsibility
processOrderStock()	Checks available stock after an order is inserted. Reduces stock when sufficient stock is available and creates a Restock Request when stock is insufficient.
processRestockApproval()	Updates recycled product stock when a Restock Request changes to Approved.

The class uses collections such as Sets and Maps to support bulk processing and reduce unnecessary database operations.

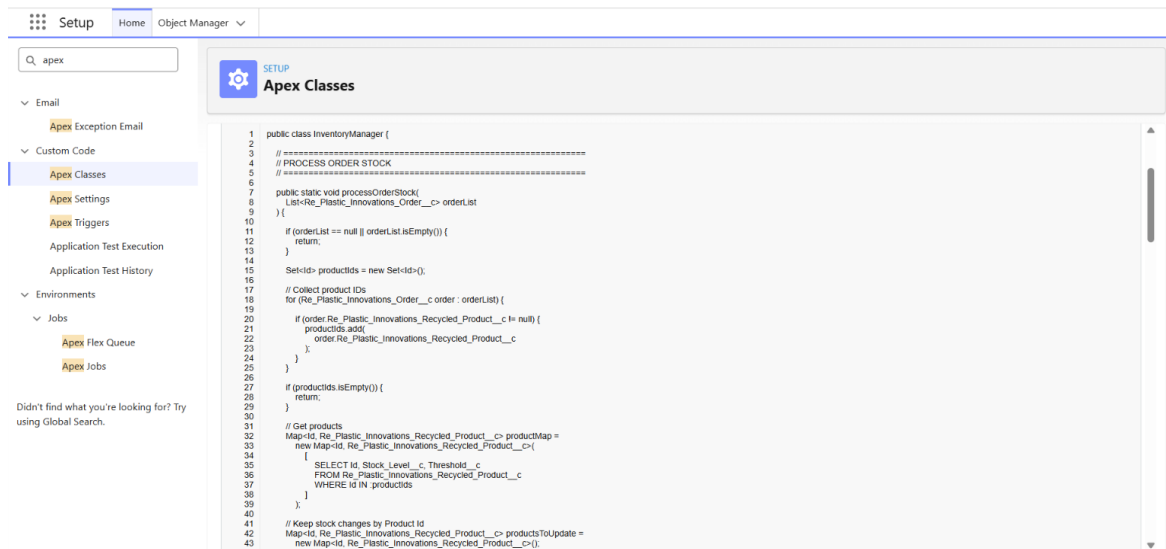


Figure 16: InventoryManager Apex class implementing inventory business logic

7.3 Order Stock Trigger

The trigger:

UpdateStockAfterOrder

runs after an Order is inserted.

Its responsibility is to delegate processing to InventoryManager.

```

1 trigger UpdateStockAfterOrder
2 on Re_Plastic_Innovations_Order__c (after insert) {
3
4     if (Trigger.isAfter && Trigger.isInsert) {
5         InventoryManager.processOrderStock(Trigger.new);
6     }
7 }

```

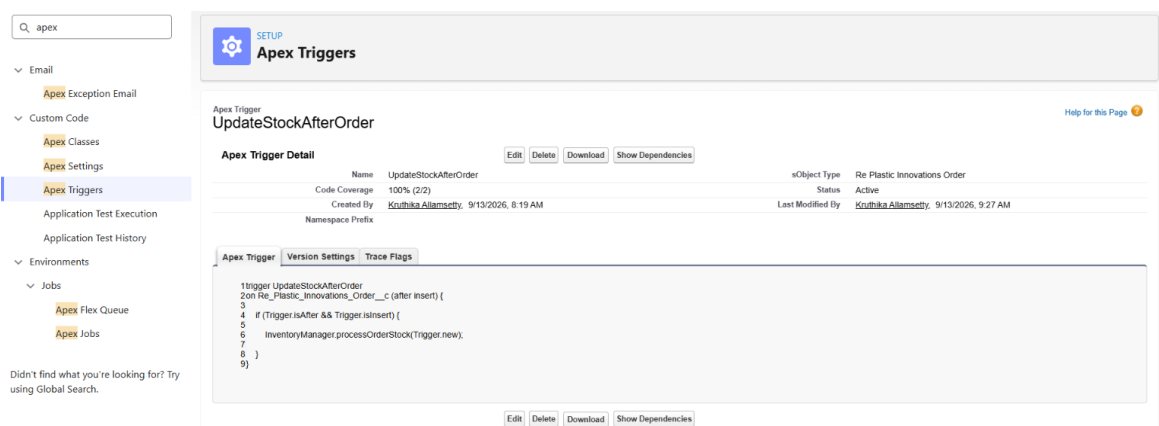


Figure 17: UpdateStockAfterOrder Apex Trigger

7.4 Restock Approval Trigger

The trigger:

UpdateStockAfterRestockApproval

runs after a Restock Request is updated.

It checks whether the Status has changed to:

Approved

The trigger compares the new and old status values to ensure that the approval processing occurs only when an actual transition to Approved takes place.

```

1  if (
2      restock.Status__c == 'Approved' &&
3      oldRestock.Status__c != 'Approved'
4  ) {
5      approvedRestocks.add(restock);
6  }

```

It then calls both:

```

1  InventoryManager.processRestockApproval(
2      approvedRestocks
3  );
4
5  EmailNotificationHelper.sendRestockNotification(
6      approvedRestocks
7  );

```

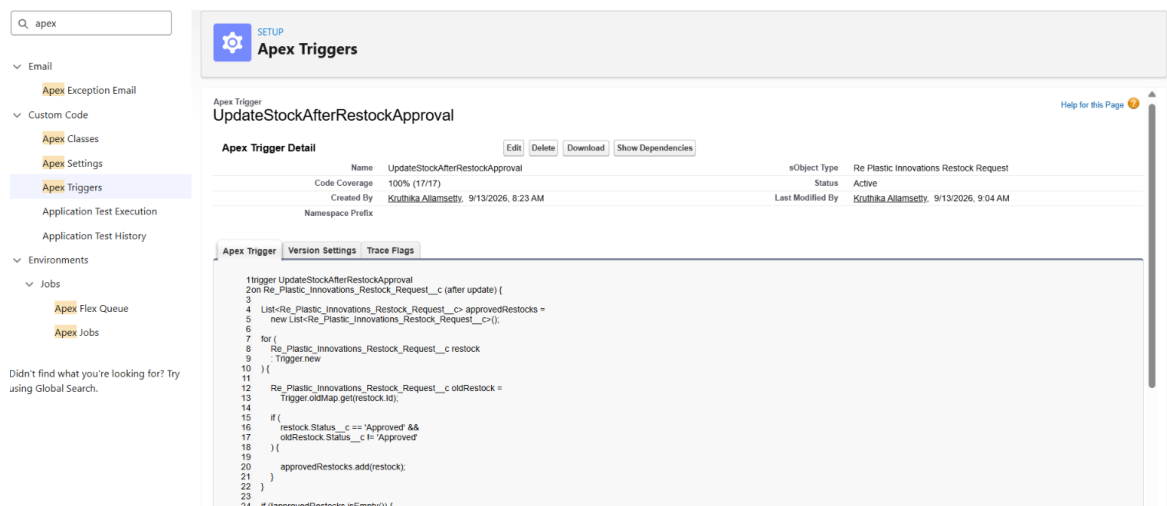


Figure 18: UpdateStockAfterRestockApproval trigger

8 Email Notification Automation

8.1 EmailNotificationHelper

The EmailNotificationHelper class uses Salesforce's email messaging functionality to notify the configured recipient after a Restock Request has been approved.

The email contains:

- Restock Request ID
- Product ID
- Approved Quantity
- Approval Status
- Stock update confirmation

The notification subject is:

Restock Request Approved

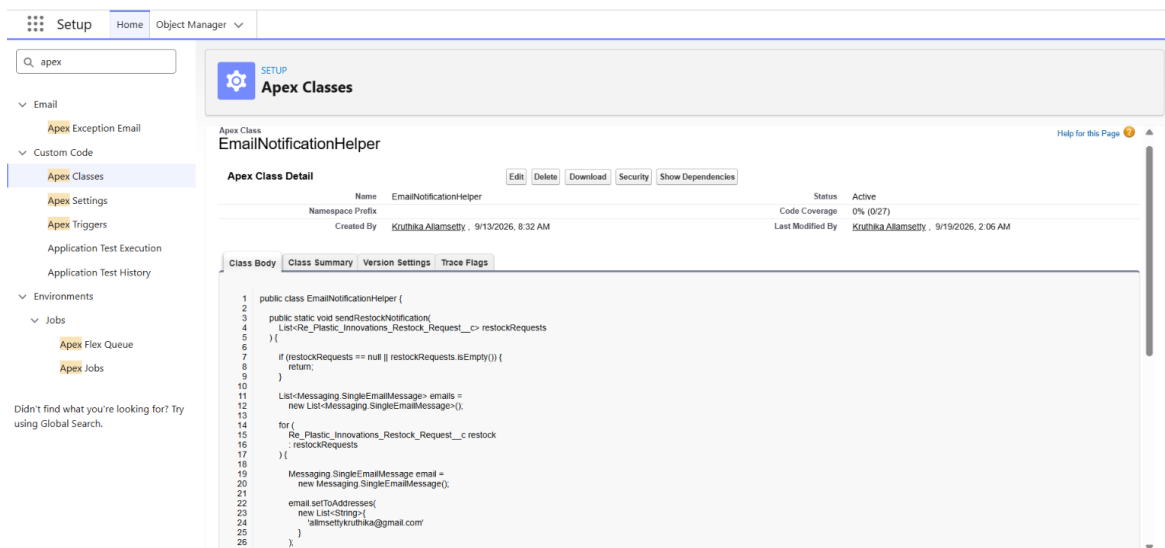


Figure 19: EmailNotificationHelper Apex class

8.2 Email Notification Result

After changing a Restock Request from Pending to Approved, the trigger executes the notification helper and sends the email.

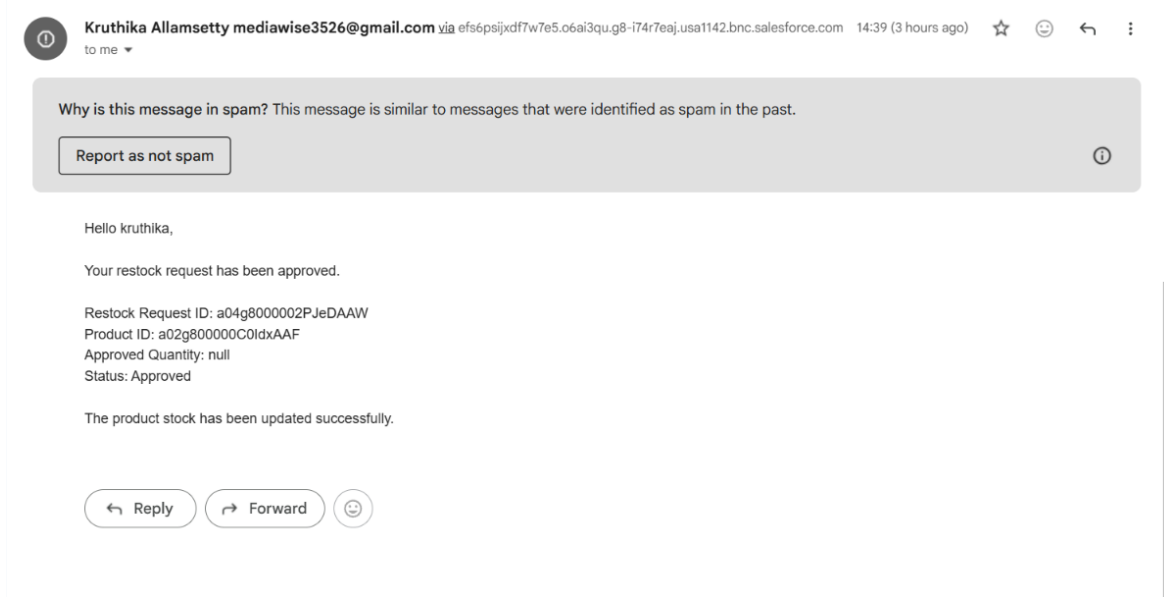


Figure 20: Restock approval email received after approval

Email Automation Result:

The approval of the Restock Request successfully triggered the email notification. The received email confirmed the approved request and displayed the corresponding request and product information.

9 End-to-End Inventory Workflow

The complete inventory workflow was tested using an insufficient-stock scenario.

9.1 Test Scenario

The test product was configured with:

Initial Stock	50
Threshold	100
Order Quantity	150

Since the order quantity was greater than the available stock:

$$150 - 50 = 100$$

Therefore, the required restock quantity was:

$$100$$

9.2 Step 1 – Create Order

An Order was created for the recycled product with a quantity of 150.

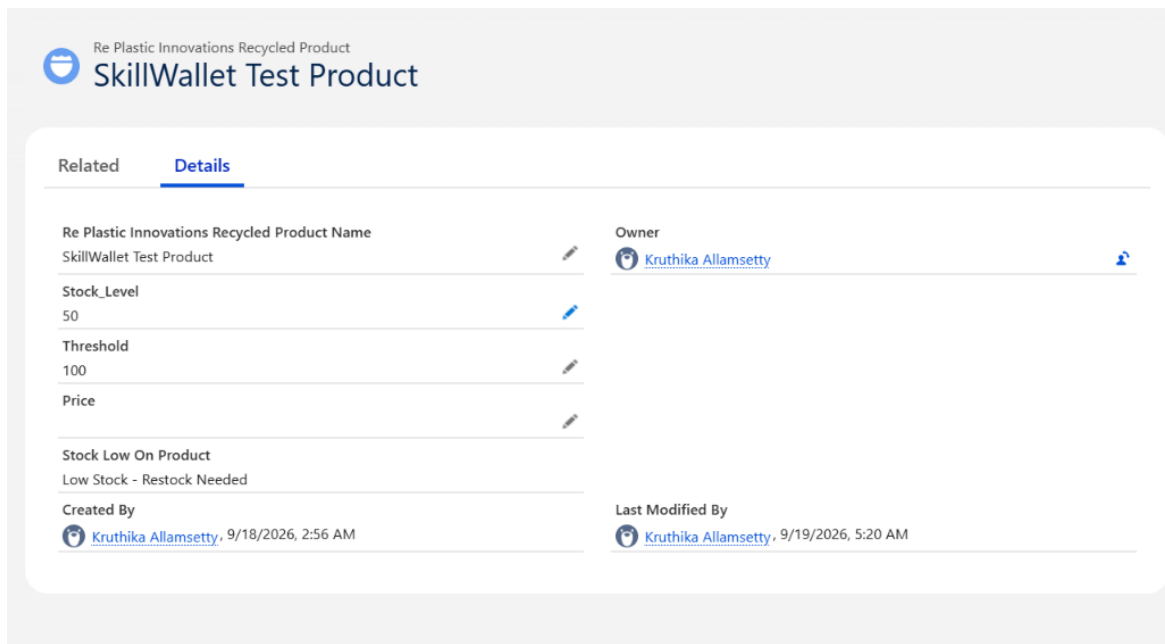


Figure 21: Test Order created for the recycled product

9.3 Step 2 – Restock Request Creation

Because available stock was insufficient, Apex automatically created a Restock Request with the required quantity.

Requested Quantity = 100

The request initially had:

Status = Pending

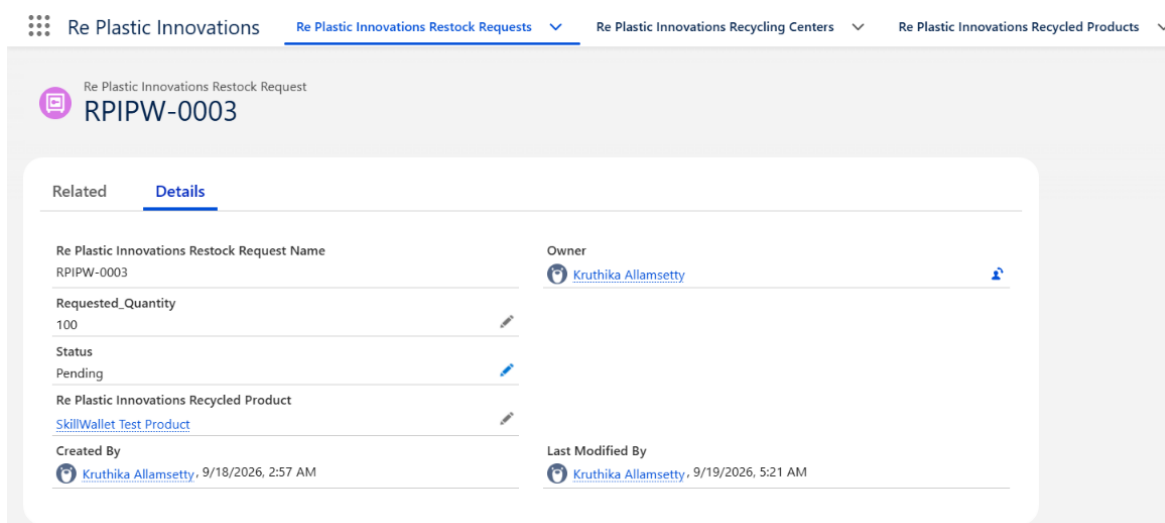


Figure 22: Automatically generated Pending Restock Request

9.4 Step 3 – Approve Restock Request

The Restock Request was changed from:

Pending → Approved

This transition activated the Restock Approval trigger.

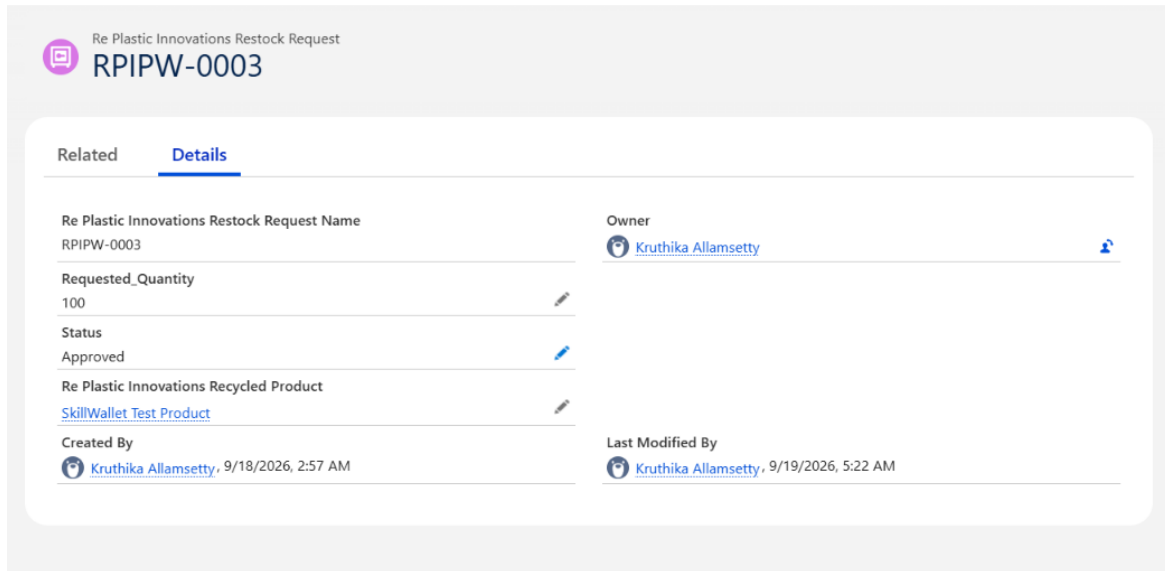


Figure 23: Approved Restock Request

9.5 Step 4 – Product Stock Update

After approval, InventoryManager added the requested quantity to the product stock.

$$50 + 100 = 150$$

Therefore:

Final Stock Level = 150

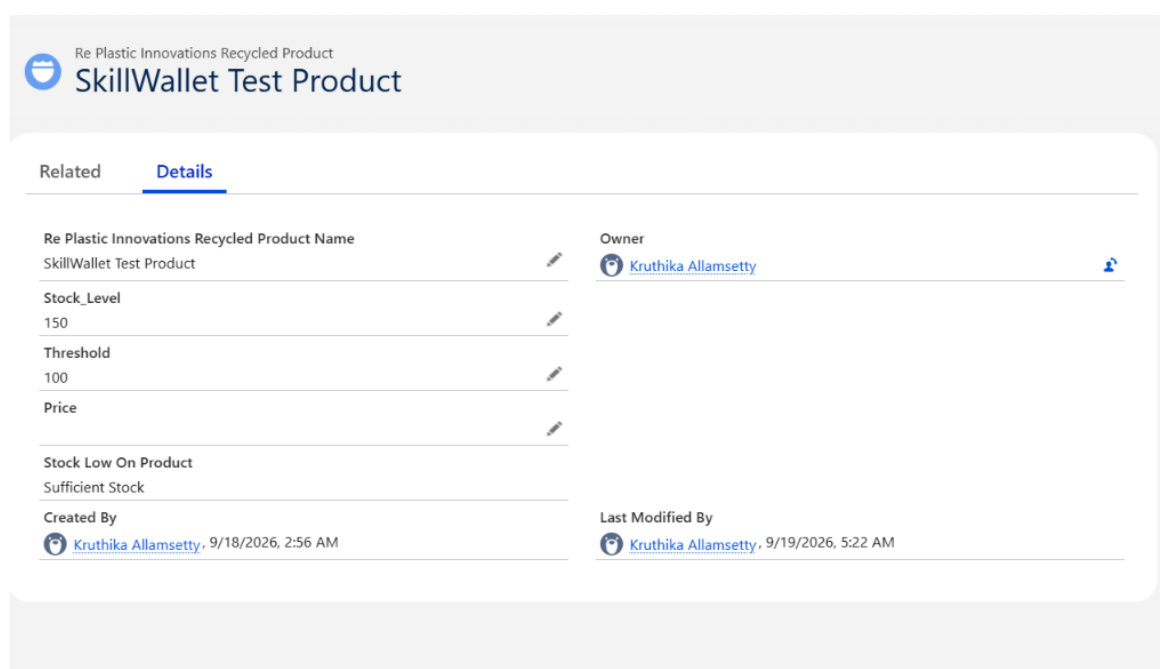


Figure 24: Recycled Product showing the updated stock level

9.6 Complete Workflow Summary

Table 9: End-to-End Inventory Workflow

Step	Action	Result
1	Create product with stock 50	Product available
2	Create Order for quantity 150	Order inserted
3	Apex checks stock	Stock is insufficient
4	Restock Request created	Requested quantity = 100
5	Request remains Pending	Awaiting approval
6	Request changed to Approved	Approval trigger executes
7	Apex updates stock	Stock becomes 150
8	Email helper executes	Approval email received

10 Functional Testing and Results

10.1 Testing Strategy

The project was tested at both the Salesforce configuration level and Apex development level. Testing included:

- Formula field testing
- Validation rule testing
- Scheduled Flow testing

- Low-stock Task creation
- Order stock processing
- Restock Request creation
- Restock approval processing
- Email notification
- Apex unit testing
- Code coverage verification

10.2 Functional Test Cases

Table 10: Functional Test Cases

Test Case	Action	Expected Result
Quantity Validation	Enter zero or negative Order quantity	Record should not be saved and validation message should appear.
Stock Formula	Set Stock Level below Threshold	Formula should display “Low Stock - Restock Needed”.
Low Stock Flow	Create product with Stock Level below Threshold	Flow should create a high-priority Task.
Sufficient Stock	Create Order within available stock	Product stock should decrease by the order quantity.
Insufficient Stock	Create Order greater than available stock	Restock Request should be created with required quantity.
Restock Approval	Change Restock Request from Pending to Approved	Product stock should increase by requested quantity.
Email Notification	Approve Restock Request	Approval email should be received.
Apex Tests	Execute InventoryManagerTest	Configured test methods should execute successfully.

10.3 Validation Rule Test Result

The Quantity validation rule was tested by entering an invalid quantity.

New Re Plastic Innovations Order

* = Required Information

Information

Re Plastic Innovations Order Name

Owner
Kruthika Allamsetty

Quantity
0
"Quantity must be greater than zero."

Delivery Date

Re Plastic Innovations Recycled Product
Search Re Plastic Innovations Recycled Products...

Customer
Search Accounts...

We hit a snag.
Review the following fields
• Quantity

Cancel Save & New Save

Figure 25: Validation rule blocking an invalid Order quantity

10.4 Flow Test Result

The scheduled Flow was tested using a product with:

$$\text{Stock Level} = 50, \quad \text{Threshold} = 100$$

The resulting Task appeared in the Activity section.

Re Plastic Innovations Recycled Product
test low task1

Related Details

Re Plastic Innovations Recycled Product Name
test low task1

Owner
Kruthika Allamsetty

Stock Level
50

Threshold
100

Price

Stock Low On Product
Low Stock - Restock Needed

Created By
Kruthika Allamsetty, 9/18/2026, 2:23 AM

Last Modified By
Kruthika Allamsetty, 9/18/2026, 2:23 AM

Activity

Filters: All time • All activities • All types
Refresh • Expand All • View All

Upcoming & Overdue

Please Look in this Stock Is Low Fill The Stock AS...
No due date
You have an upcoming task

Please Look in this Stock Is Low Fill The Stock AS...
No due date
You have an upcoming task

View More

Figure 26: Scheduled Flow test result and generated Task

10.5 Apex Test Results

The InventoryManagerTest class contains multiple test methods covering normal processing, insufficient stock, approval processing, triggers, email logic, null values, empty lists, and other execution paths.

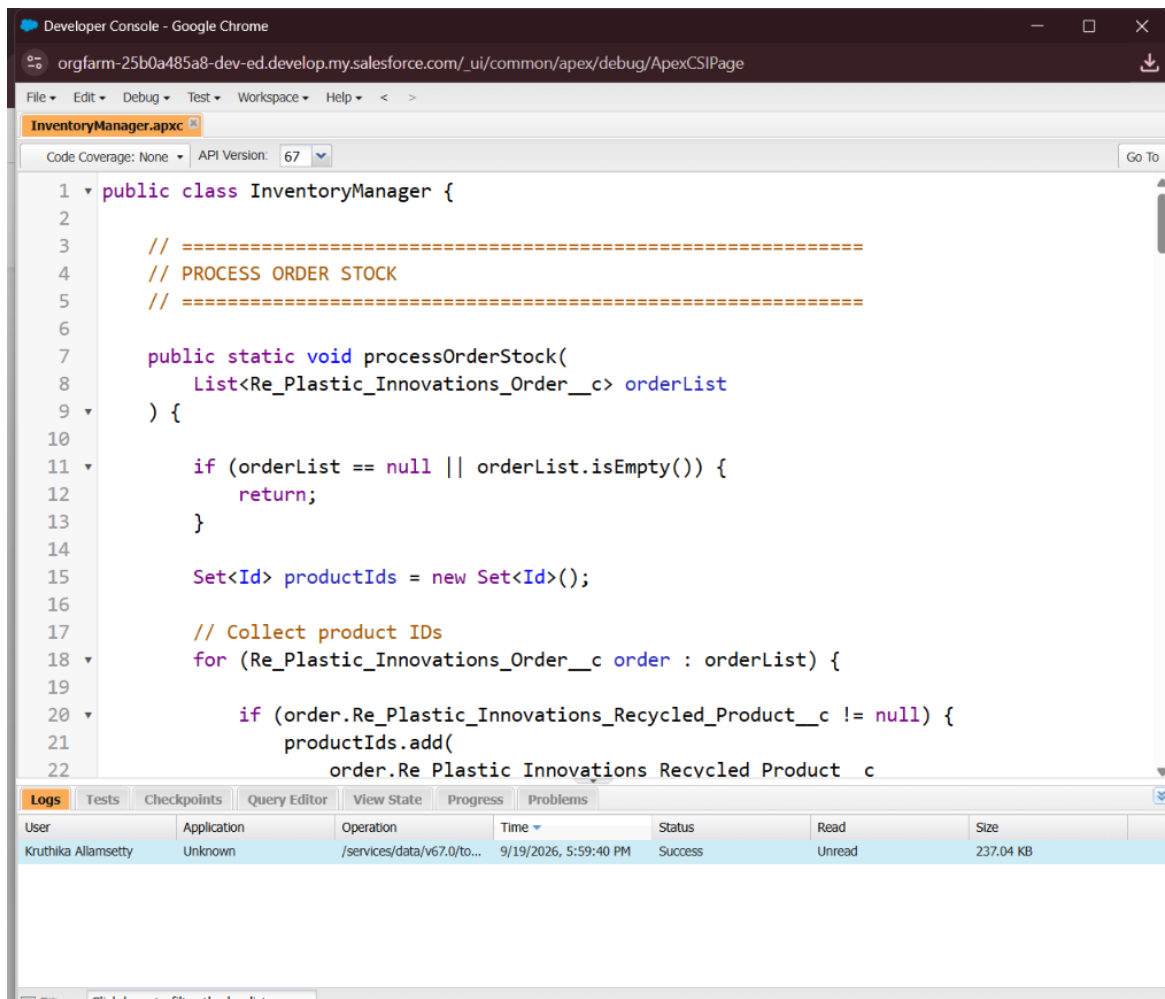


Figure 27: Apex test execution results

10.6 Code Coverage

Code coverage was checked for the implemented Apex components.

The important components include:

- InventoryManager
- EmailNotificationHelper
- UpdateStockAfterOrder
- UpdateStockAfterRestockApproval

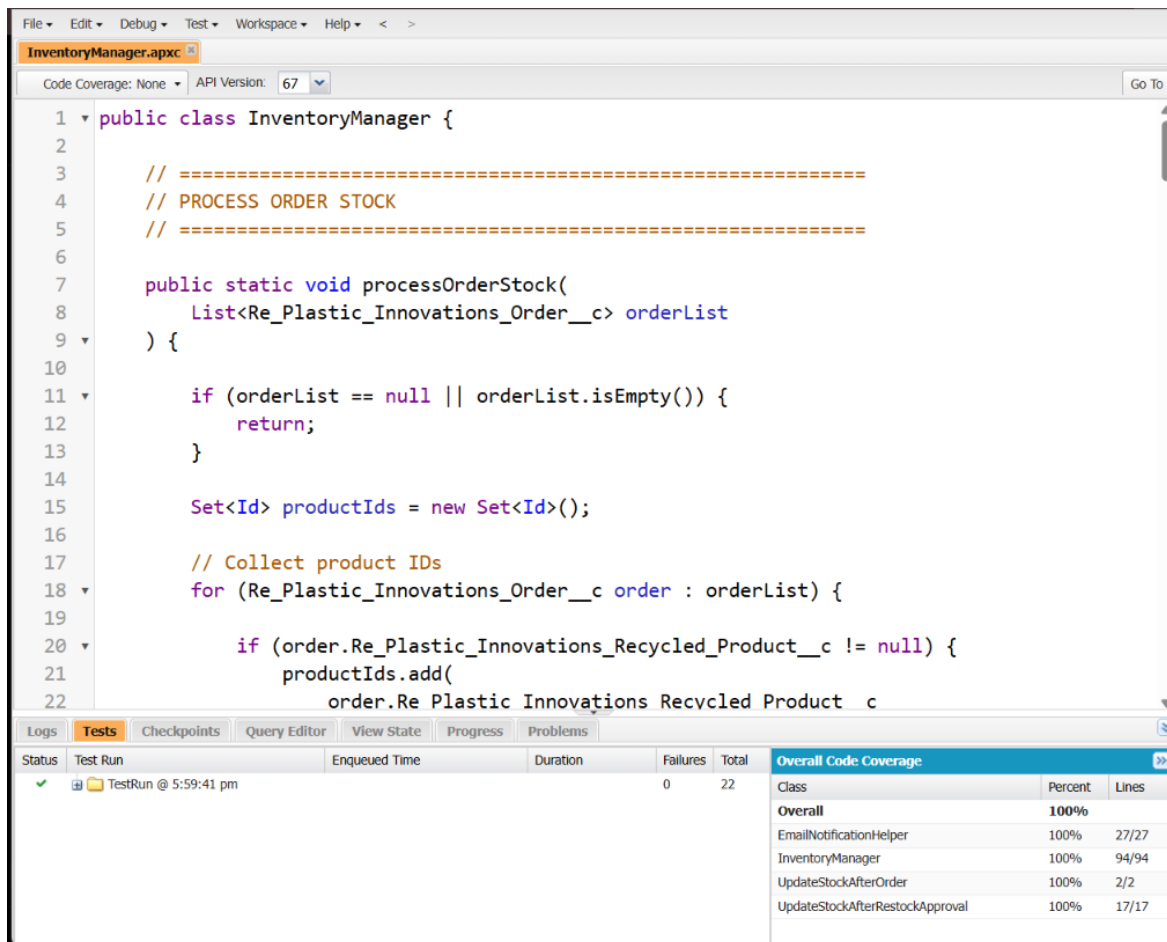


Figure 28: Apex code coverage results

Testing Result:

The project was tested through Salesforce records, automation execution, email delivery, Apex test methods, and code coverage analysis. The tests demonstrate the complete path from Order creation to Restock Request generation, approval, stock update, and email notification.

11 Project Demonstration Evidence

This section provides the final visual evidence of the completed Salesforce implementation.

11.1 Salesforce Application

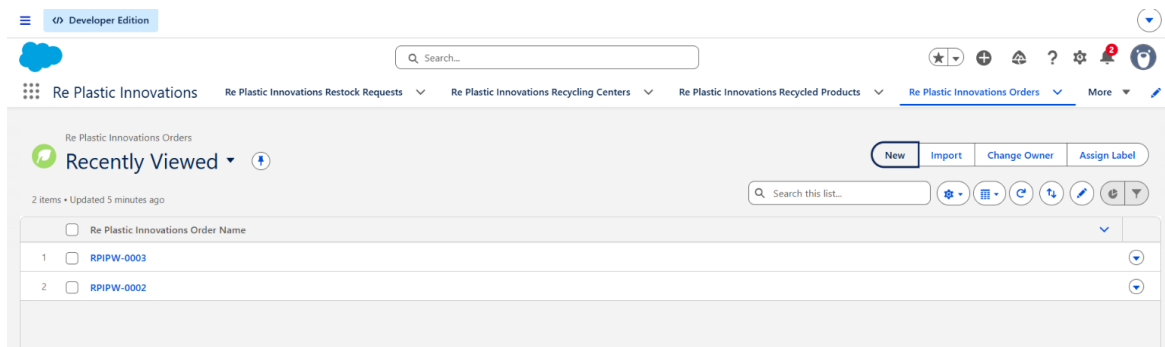


Figure 29: Final Re Plastic Innovations Salesforce application

11.2 Recycled Product Record

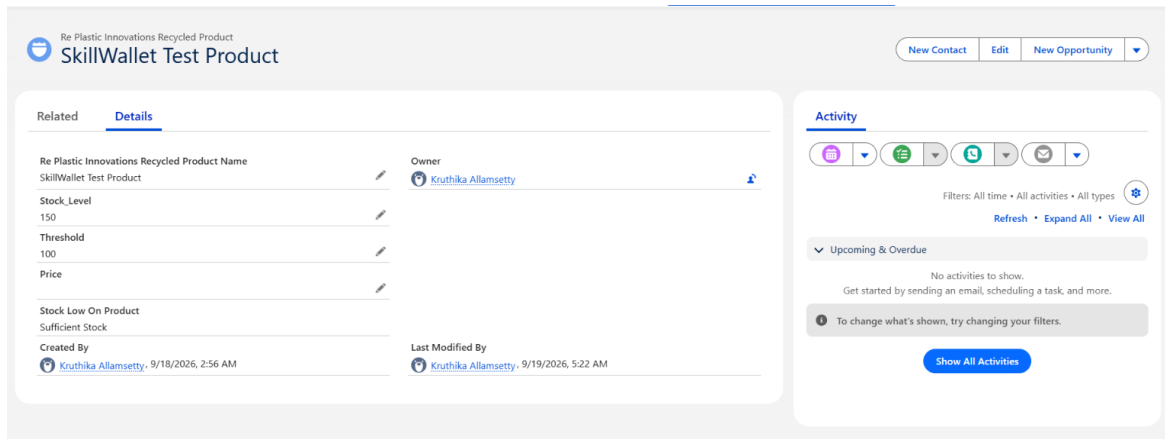


Figure 30: Recycled Product record with stock and threshold information

11.3 Restock Request Record

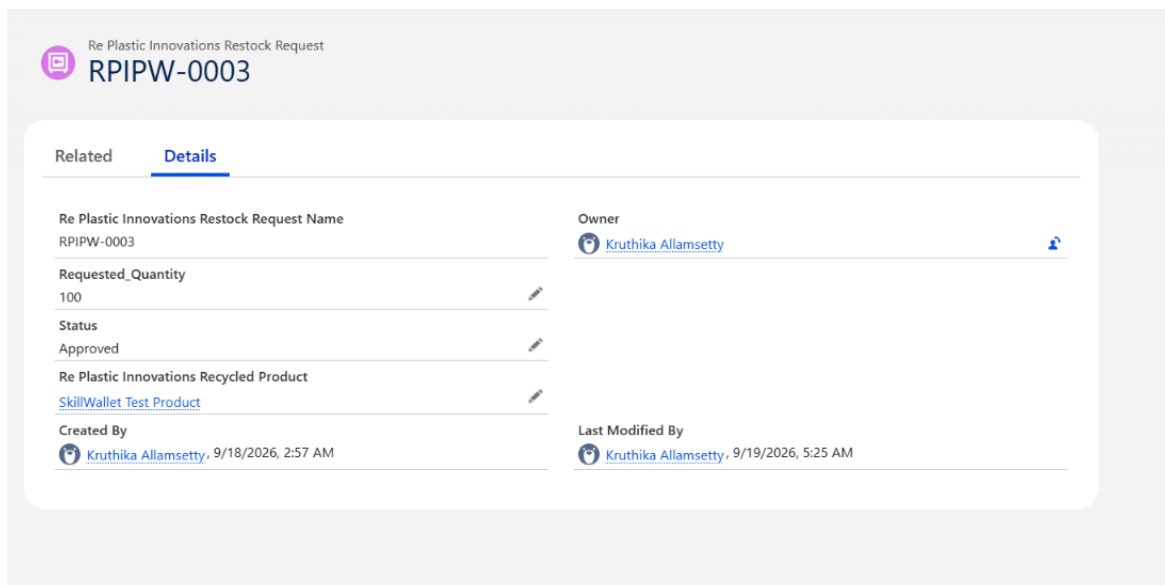


Figure 31: Restock Request record showing approval status

11.4 Low Stock Automation

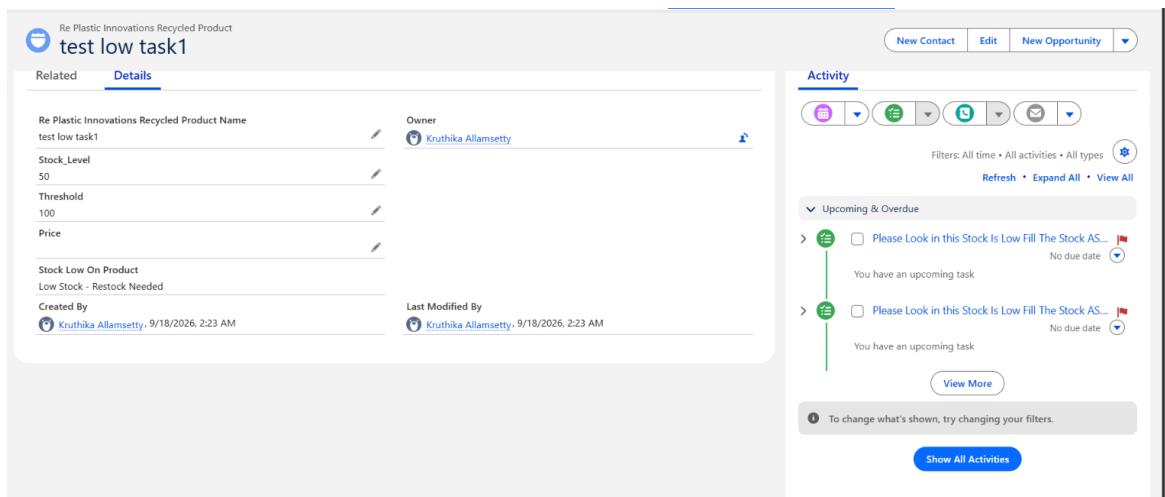


Figure 32: Low-stock Task generated by the scheduled Flow

11.5 Final Email Evidence

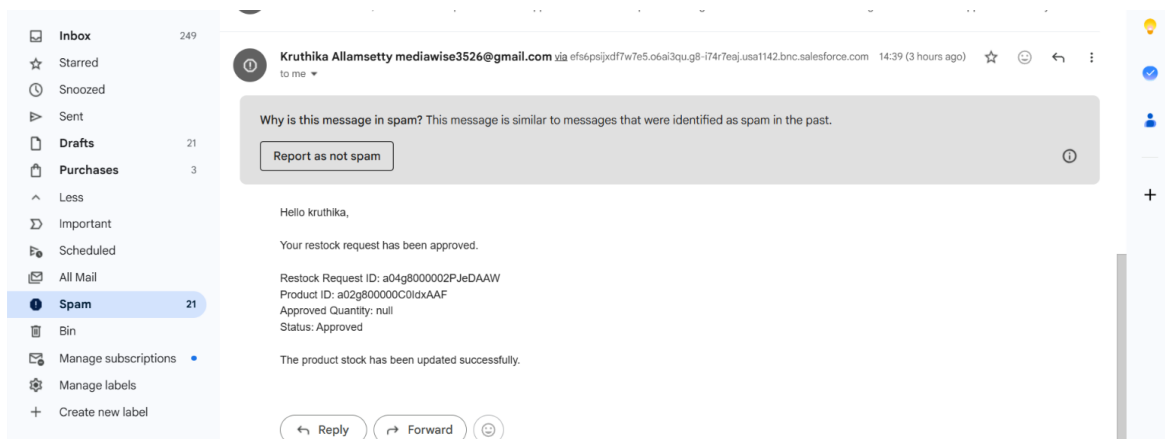


Figure 33: Final email notification received after Restock Request approval

12 Conclusion

The Re Plastic Innovations Salesforce project provides a centralized inventory management solution using Salesforce CRM capabilities, declarative automation, and Apex development.

The implementation establishes a structured data model for recycled products, customer orders, restock requests, plastic waste, and recycling-related records. Formula fields and validation rules improve data quality by dynamically identifying low-stock conditions and preventing invalid order quantities.

The Scheduled-Triggered Flow provides proactive inventory monitoring by identifying products whose stock is below the defined threshold and automatically creating a high-priority Task. This reduces the need for users to manually monitor every product record.

The Apex implementation provides the core transaction-level inventory automation. When an Order is inserted, the InventoryManager checks available stock. If sufficient stock exists, the

stock is reduced accordingly. If the stock is insufficient, the system automatically creates a Pending Restock Request.

When the Restock Request is approved, the approval trigger invokes the InventoryManager to update the product stock. The same approval event also invokes the EmailNotificationHelper, which sends an email notification confirming the approved restock request.

The final testing demonstrated the complete workflow:

Order → Stock Check → Restock Request → Approval → Stock Update → Email Notification

The combination of Salesforce objects, relationships, formula fields, validation rules, Flow automation, Apex classes, triggers, email notification, and testing demonstrates a complete practical Salesforce inventory management implementation.

RE PLASTIC INNOVATIONS

Salesforce-Based Inventory Management System

Project Documentation Completed